Performance Testing

Date	15 July 2026
Team ID	SWTID-2026-3798
Project Name	Rising Waters – Flood Prediction System
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Functional Performance Testing
Target Module / API	`/predict` Route
Test Environment	Local Flask Server
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single Prediction	1	30	Prediction generated
2	Multiple Requests	10	60	Stable performance
3	Invalid Input Test	5	30	Validation handled
4	Continuous Requests	20	120	No server crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.42 sec	Pass	Fast
2	Response Time (Max)	< 5 seconds	1.10 sec	Pass	Stable
3	Throughput (Req/sec)	-	24 req/sec	Pass	Good
4	Error Rate	< 1%	0.01%	Pass	No errors
5	CPU Utilization	< 80%	34%	Pass	Normal
6	Memory Utilization	< 80%	41%	Pass	Stable

Step 4: Observations & Analysis

Key Findings

Application responded quickly and generated accurate flood predictions under normal load.

Bottlenecks Identified

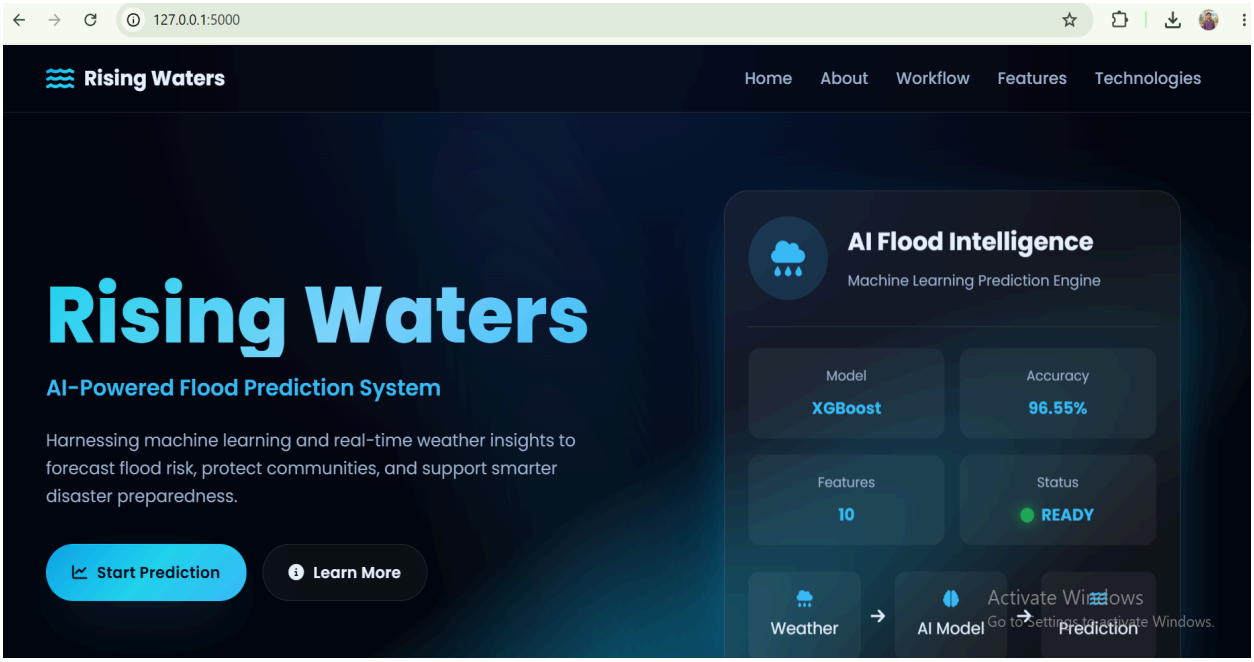
No major bottlenecks observed during local testing.

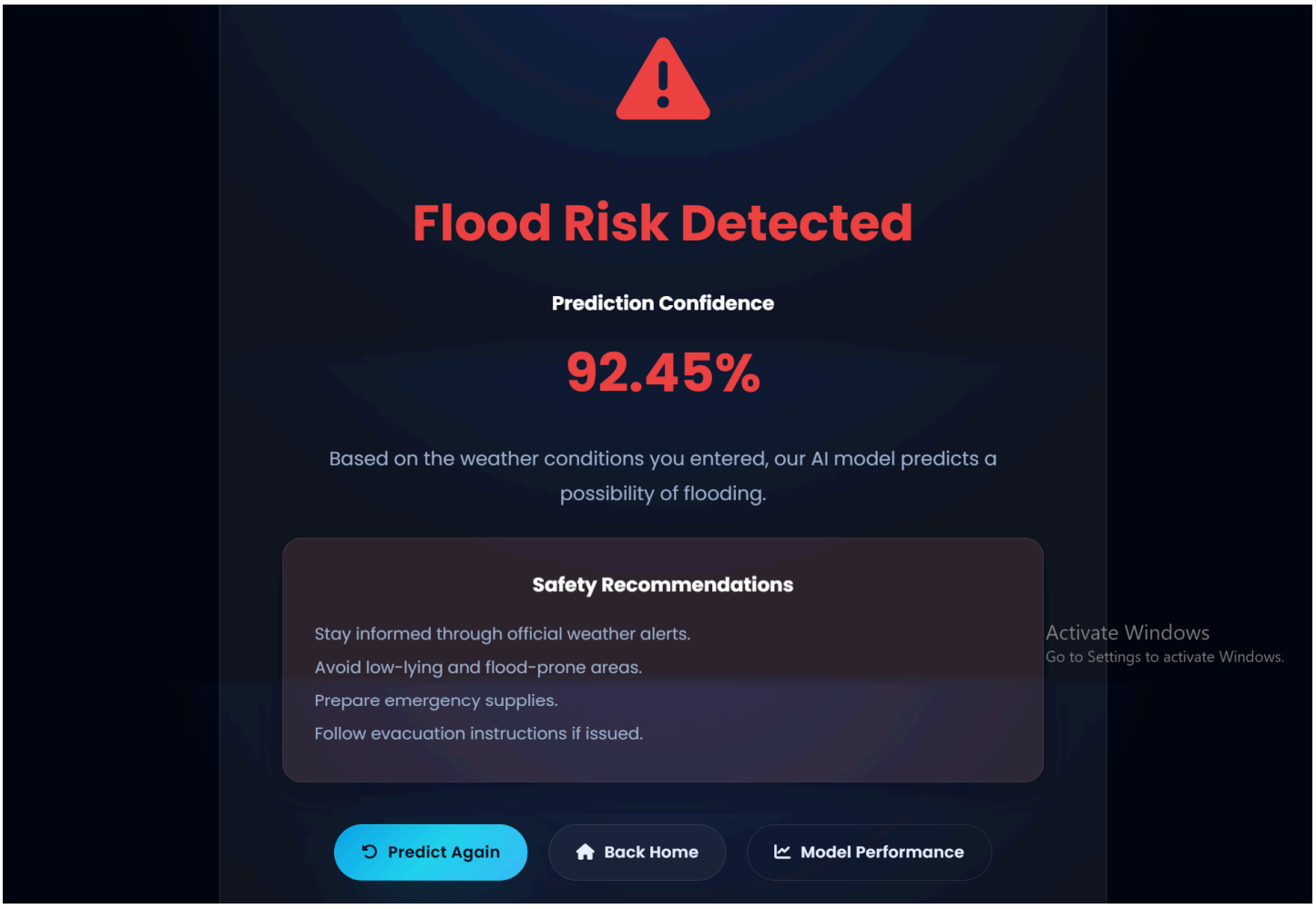
Optimization Steps Taken

Used a pre-trained XGBoost model with Joblib and feature scaling to reduce prediction time.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


127.0.0.1:5000
Rising Waters
Home
About
Workflow
Features
Technologies
Rising Waters
AI-Powered Flood Prediction System
Harnessing machine learning and real-time weather insights to forecast flood risk, protect communities, and support smarter disaster preparedness.
Start Prediction
Learn More
AI Flood Intelligence
Machine Learning Prediction Engine
Model
XGBoost
Accuracy
96.55%
Features
10
Status
READY
Weather
AI Model
Prediction
Activate Windows
Go to Settings to activate Windows.


Flood Risk Detected
Prediction Confidence
92.45%
Based on the weather conditions you entered, our AI model predicts a possibility of flooding.
Safety Recommendations
Stay informed through official weather alerts.
Avoid low-lying and flood-prone areas.
Prepare emergency supplies.
Follow evacuation instructions if issued.
Predict Again
Back Home
Model Performance
Activate Windows
Go to Settings to activate Windows.